Emotional Storyteller for Vision Impaired and Hearing-Impaired Children.

(EMOTIONALLY EXPRESSIVE READER FOR CHILDREN WITH SENSORY CHALLENGES)

STORYPAL — TEXT-TO-SPEECH WITH EMOTION SYNTHESIS

Project Final Report

E. Sentheepan – IT22282590

B.Sc. (Hons) in Information Technology Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology
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DECLARATION

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The above candidate is carrying out research for the undergraduate dissertation under my supervision.

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Signature of the supervisor	Date
(Ms. Sanjeevi Chandrasiri)	

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Signature of co-supervisor	Date
(Ms. Karthiga Rajendran)	

ABSTRACT

In an increasingly digital world, children with visual and hearing impairments face significant barriers in accessing the richness of storytelling experiences. This research focuses on one of four key modules within Storypal — a mobile application designed for inclusive, emotionally enriched storytelling: the Conversion of Text to Human Natural Audio with Emotion. This module transforms text extracted from storybooks into emotionally expressive synthesized speech, making narratives accessible and engaging for visually impaired children.

The proposed system employs Tacotron2, a sequence-to-sequence neural network, to convert text into Mel spectrograms, which are then converted into high-quality waveforms using WaveGlow — a flow-based generative model. A reference encoder is additionally integrated into the pipeline to facilitate emotional style emulation, enabling the synthesized audio to reflect the detected emotional tone of each sentence.

The emotion detection component, powered by BERT (Bidirectional Encoder Representations from Transformers), classifies each sentence's emotional tone into seven categories: joy, sadness, anger, fear, disgust, surprise, and neutral. The detected emotion is then fed into the reference encoder to modulate the speaking style of the generated audio.

Evaluation of the system shows that the neutral text-to-speech synthesis achieves a Mean Opinion Score (MOS) of 3.65 ± 0.05, approaching human-level speech quality. The emotional synthesis, while achieving a lower MOS of 1.65 ± 0.05, demonstrates the feasibility of the approach and highlights clear directions for future improvement. This research contributes to the growing field of emotionally expressive speech synthesis and affirms the potential of deep learning models to make storytelling truly inclusive.

Keywords: Text-to-Speech, Tacotron2, WaveGlow, Emotional Speech Synthesis, BERT, Emotion Detection, Inclusive Technology, Storypal, Neural TTS, Mel Spectrogram

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
TTS	Text-to-Speech
NLP	Natural Language Processing
BERT	Bidirectional Encoder Representations from Transformers
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory
CTC	Connectionist Temporal Classification
MOS	Mean Opinion Score
API	Application Programming Interface
ASR	Automatic Speech Recognition
OCR	Optical Character Recognition
GPU	Graphics Processing Unit
SLIIT	Sri Lanka Institute of Information Technology
SUS	System Usability Scale
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
ISEAR	International Survey on Emotion Antecedents and Reactions
ASL	American Sign Language
WER	Word Error Rate
LJSpeech	LJ Speech Dataset

I. INTRODUCTION

The act of storytelling holds immense significance in a child's life. It transcends the boundaries of culture, language, and time, fostering imagination, creativity, and cognitive development. Stories are not just narratives; they are windows to different worlds, vehicles of moral lessons, and sparks of inspiration. However, in our fast-paced, technology-driven world, a pressing issue has emerged: children's diminishing access to the enriching world of stories. While the problem itself is multi-faceted, encompassing aspects of time constraints for parents and the evolving reading habits of children, the issue is exacerbated for visually impaired children who face additional barriers to accessing conventional storybooks that are not available in Braille.

Visually impaired children often struggle to access the rich visual content of storybooks. Traditional books rely heavily on illustrations to convey the narrative, leaving visually impaired children with a limited understanding of the story's visual elements. On the other hand, hearing-impaired children may miss out on the auditory aspects of storytelling, including the nuances of tone, pitch, and emotion conveyed through spoken words. The simple pleasure of a bedtime tale becomes an inaccessible dream for these children.

The development of inclusive storytelling platforms, like Storypal, signifies a broader trend toward making literature and educational content accessible to all. These platforms are not limited to children but also extend their benefits to adults with disabilities. The integration of technology, including artificial intelligence and machine learning, is at the forefront of this inclusive storytelling revolution.

Storypal is a collaborative research initiative comprising four interconnected modules: (1) Text Extraction from storybook images, (2) Emotion Detection at the sentence level, (3) Text-to-Speech Synthesis with Emotional Expression, and (4) Text-to-Sign Language Translation. This report focuses exclusively on the third module — the Text-to-Speech (TTS) Synthesis component — developed by Sentheepan. E (IT22282590).

The Text-to-Speech module is the auditory bridge of Storypal. It takes the structured, emotion-tagged text output from the OCR and emotion detection modules and synthesizes natural-sounding, emotionally expressive speech. This speech output narrates the story to visually impaired children in a manner that preserves the emotional context of each sentence — happiness, sadness, anger, fear, disgust, surprise, or neutrality — making the listening experience both informative and emotionally resonant.

The core engine of this module is Tacotron2, an end-to-end sequence-to-sequence model that converts text characters directly into Mel spectrograms. WaveGlow, a flow-based generative vocoder, then converts these spectrograms into high-quality audio waveforms. An additional reference encoder is incorporated to guide the emotional speaking style, allowing the synthesized voice to adapt to the emotional tone detected by the BERT-based classifier.

In conclusion, this component of Storypal represents a significant stride in the pursuit of accessibility and inclusive education. By transforming plain text into emotionally rich audio narratives, this research aims to create a world where all children — regardless of their visual or hearing abilities — can experience the magic of storytelling. This work builds upon existing literature in speech synthesis, emotional computing, natural language processing, and assistive technology to pave the way for a more inclusive future in children's literature.

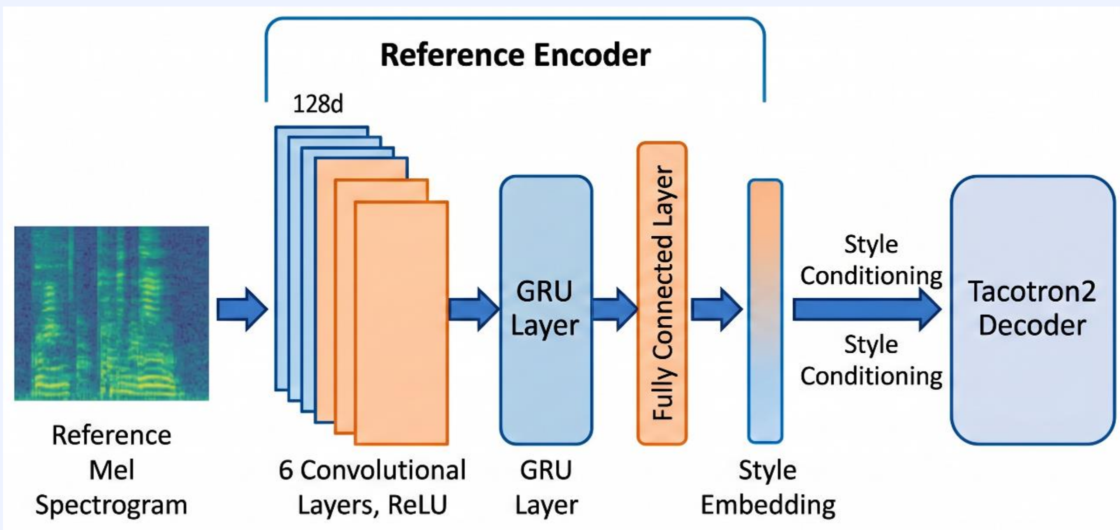


Figure 1: Reference Encoder for-emotionaltyle trrnaitt of emotional style traunaiitation ifigured pin the neuralenocorr emotion2 decoder in TTS.

1.1 Background Study and Literature Review

1.1.1 Background Study

Text-to-Speech (TTS) synthesis has undergone a remarkable transformation over the past decade, shifting from rule-based systems that produced robotic, monotonous output to deep learning models that generate speech nearly indistinguishable from human voice. The evolution has been driven largely by advances in neural network architectures, particularly sequence-to-sequence models and generative networks.

Early TTS systems such as Festival and eSpeak relied on concatenative synthesis or formant synthesis. These systems, while functional, lacked naturalness and were incapable of conveying emotional nuance. The introduction of WaveNet in 2016 by DeepMind marked a turning point: using dilated causal convolutions, WaveNet could generate raw audio waveforms at the sample level, achieving unprecedented speech quality. However, its autoregressive nature made it computationally expensive and slow for real-time applications.

Tacotron (2017) addressed the need for an end-to-end model by learning to map character sequences directly to spectrograms, removing the need for hand-crafted acoustic features. Its successor, Tacotron2 (2018), combined a recurrent sequence-to-sequence network for spectrogram prediction with WaveNet as a vocoder, achieving human-level speech synthesis on standard benchmarks.

Despite these advances, the integration of emotion into synthesized speech remained a significant challenge. Natural human speech is not tonally neutral; it carries subtle and overt emotional cues that profoundly affect listener comprehension and engagement. For applications in storytelling and education — particularly for children — neutral speech synthesis is insufficient.

Research into emotional TTS has explored style transfer mechanisms, reference encoders, and generative adversarial networks. Notable works include the Global Style Token (GST) architecture by Wang et al. (2018), which introduced learnable tokens representing different speaking styles, and the Mellotron model by Valle et al. (2020), which allowed fine-grained control over pitch, rhythm, and style.

In the context of children with visual impairments, the need for emotionally expressive TTS is particularly acute. When a child cannot see the illustrations or the facial expressions of a reader, the emotional content of the story must be conveyed entirely through audio. A monotone narrative fails to capture the excitement of an adventure, the sadness of a loss, or the joy of reunion — critical elements that support comprehension, empathy development, and engagement.

This research builds upon the Tacotron2 + WaveGlow pipeline and extends it with a reference encoder mechanism to achieve emotion-conditioned speech synthesis. The broader context of this work is the Storypal application — a mobile platform designed to make storybooks accessible to visually and hearing-impaired children through multimodal AI technologies.

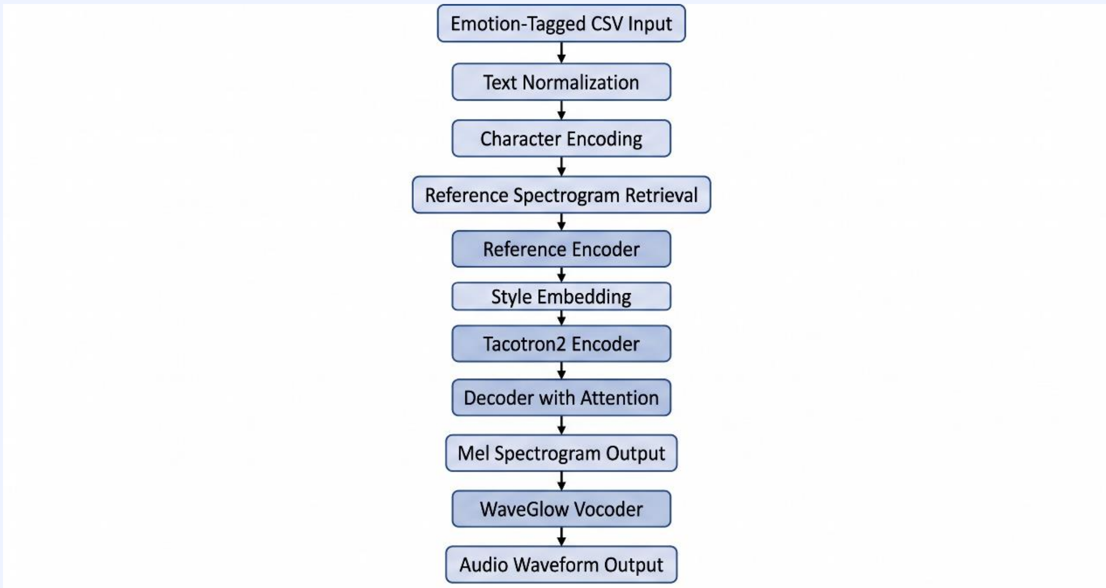


Figure 17. Text-oto-eech synthesis algorithm speech wetleris.

1.2 Literature Review

The landscape of text-to-speech synthesis and its intersection with emotion-aware computing has been the subject of extensive academic investigation. This section reviews the most relevant studies that inform the design and implementation of the TTS module within Storypal.

The foundational work in neural TTS was established by Van den Oord et al. (2016) with WaveNet, a deep autoregressive model capable of generating raw waveforms from acoustic features. While achieving exceptional audio quality, WaveNet's sequential generation was too slow for real-time deployment. This limitation was later addressed by Prenger et al. (2019) who introduced WaveGlow — a flow-based generative network that parallelizes inference by combining elements of the Glow model with WaveNet's signal processing insights. WaveGlow eliminates the need for auto-regression while maintaining high-quality synthesis, making it suitable for production deployment.

Shen et al. (2018) proposed Tacotron2, a landmark contribution to end-to-end TTS research. The model uses a convolutional-recurrent encoder, a location-sensitive attention mechanism, and an autoregressive decoder to produce Mel spectrograms from character sequences. When paired with a modified WaveNet vocoder, Tacotron2 achieved a MOS of 4.53 — close to the natural speech MOS of 4.58. This benchmark established Tacotron2 as the state-of-the-art foundation for subsequent TTS research.

Wang et al. (2018) introduced the Global Style Token (GST) framework as an extension of Tacotron, allowing the model to learn unsupervised speaking style representations. These style embeddings can be combined with text representations to control the prosody and emotional quality of synthesized speech without explicit emotion labels. This work directly informs the reference encoder design used in Storypal's TTS module.

For emotion-aware synthesis, Tuvshinjargal et al. (2024) developed SRC-IT2, a speech-rate-controllable emotional TTS system based on an improved Tacotron2 architecture. Their system demonstrated that explicit emotional conditioning alongside prosodic control yields more natural and expressive output. Similarly, Wang and Zhang (2025) conducted a comprehensive survey of emotional TTS models, documenting the field's progression from rule-based emotion insertion to end-to-end emotionally conditioned generation.

In the domain of emotion recognition from text — a necessary precursor to emotion-conditioned TTS — Devlin et al. (2018) introduced BERT, a transformer-based language model pre-trained on vast text corpora via masked language modeling and next sentence prediction tasks. BERT's bidirectional context sensitivity makes it highly effective for downstream tasks including sentiment analysis and emotion classification. Pratama and Kurniawan (2026) extended BERT's applicability with a Fuzzy-BERT framework, demonstrating improved performance on multi-label emotion classification for narrative text. Hossain et al. (2025) further validated the use of BERT with BiLSTM for emotion sensing in healthcare contexts, reaffirming BERT's versatility.

In the field of inclusive technology for children with disabilities, Roseline et al. (2025) developed an inclusive learning platform for visual and hearing-impaired students, noting the critical importance of multimodal access — audio, visual, and haptic — in supporting diverse learning needs. Miller and Thompson (2025) proposed NiteStory.AI, an AI-powered tool for generating personalized social stories for children, demonstrating the practical viability of NLP and TTS in educational accessibility tools.

The intersection of these research threads — neural TTS, emotional speech synthesis, BERT-based emotion detection, and inclusive education technology — forms the theoretical and technical foundation of this research. By synthesizing insights from these works, the Storypal TTS module aims to advance the state of accessible, emotionally expressive storytelling technology for children with visual impairments.

1.3 Research Gap

Identifying research gaps is crucial for advancing inclusive education and ensuring that new studies address the unmet needs of children with sensory challenges. While significant progress has been made in both neural text-to-speech synthesis and emotion-aware computing, a critical intersection remains insufficiently explored: the development of real-time, emotionally expressive TTS systems designed specifically for the context of children's storytelling and accessible education.

The following table summarizes the key gaps identified through review of existing literature:

Study	Contribution	Limitation	Gap Addressed by This Research
Shen et al. (2018) – Tacotron2	End-to-end neural TTS with near-human MOS	No emotional conditioning; neutral speech only	Emotional conditioning via reference encoder
Wang et al. (2018) – GST	Unsupervised style token learning for prosody control	Style control is abstract; not mapped to explicit emotions	Explicit emotion-to-style mapping from BERT classifier
Tuvshinjargal et al. (2024) – SRC-IT2	Emotion + speech-rate control in Tacotron2	Evaluated on adult speech; not optimized for children's accessibility	Child-friendly deployment in Storypal mobile app
Roseline et al. (2025)	Inclusive platform for visual/hearing-impaired students	Limited TTS integration; no emotional expressiveness discussed	Full TTS pipeline with emotional narration
Miller & Thompson (2025) – NiteStory.AI	Personalized story generation for children	Uses commercial TTS API without emotional modulation	Custom emotion-conditioned synthesis end-to-end
Hossain et al. (2025)	BERT + BiLSTM for emotion sensing	Focused on healthcare; not connected to TTS or accessibility tools	Integration of BERT emotion detection with TTS for storytelling

Table 1: Research Gap Summary

Overall, the existing literature highlights a clear gap: while studies have individually addressed neural TTS quality, prosodic style control, emotion recognition, and inclusive educational technologies, few have integrated all of these capabilities into a single, child-accessible, real-time storytelling system. This research aims to fill this gap by developing an emotion-conditioned TTS pipeline (Tacotron2 + WaveGlow + Reference Encoder) guided by BERT-based emotion detection, deployed within the Storypal application for visually impaired children.

1.4 Research Problems

The current landscape of assistive audio tools for children with visual impairments is marked by several significant limitations that undermine their educational effectiveness and accessibility.

The most pressing problem is the lack of emotional expressiveness in existing text-to-speech systems. Most commercially available and open-source TTS tools produce monotone or emotionally neutral speech. While this is adequate for conveying factual information, it is fundamentally insufficient for storytelling, where emotional tone is a primary vehicle of meaning, character development, and narrative engagement. For a visually impaired child who depends entirely on audio to access a storybook, neutral narration eliminates the emotional richness that makes stories compelling and comprehensible.

A second critical issue lies in the difficulty of automatically detecting and translating textual emotion into speech prosody. Emotion in text is expressed through lexical choices, sentence structure, and contextual cues that require deep linguistic understanding to interpret. Mapping these textual cues to corresponding prosodic modifications — changes in pitch, speaking rate, intensity, and voice quality — in real time presents a substantial technical challenge. Existing systems either rely on manual emotion tags or handle only a limited subset of emotions.

A third challenge is the computational cost of high-quality neural speech synthesis. Models such as WaveNet are known to produce exceptional audio quality but are prohibitively slow for real-time deployment on mobile devices. Finding an architecture that balances audio quality with inference efficiency is therefore a critical research problem for this module.

Furthermore, there is a lack of child-appropriate, accessible TTS systems specifically designed for educational storytelling. Most TTS research targets adult users or general-purpose applications, and system interfaces are rarely designed for child users or integrated with storybook content delivery.

These challenges highlight the urgent need for an emotionally expressive, computationally efficient, and child-accessible TTS system that can be seamlessly integrated into a mobile storytelling platform. The proposed TTS module within Storypal aims to address these problems by combining Tacotron2, WaveGlow, BERT-based emotion detection, and a reference encoder into a unified, real-time, and emotionally aware speech synthesis pipeline.

1.5 Research Objectives

Main Objective

The main objective of this research is to develop a Text-to-Speech synthesis module that converts written story text into natural-sounding, emotionally expressive speech. The system integrates Tacotron2-based spectrogram generation, WaveGlow-based vocoding, BERT-based emotion detection, and a reference encoder to produce audio narratives that faithfully capture the emotional tone of each sentence, enhancing storytelling accessibility for visually impaired children.

Specific Objectives

- Develop a Tacotron2-based TTS model capable of generating high-quality Mel spectrograms from character-level text input.
- Integrate WaveGlow as a high-fidelity, parallel vocoder to convert Mel spectrograms to audio waveforms in near real time.
- Implement a BERT-based emotion classifier using the ISEAR dataset to categorize sentence-level emotion into seven classes: joy, sadness, anger, fear, disgust, surprise, and neutral.
- Design and integrate a reference encoder that modulates the speaking style of the Tacotron2 model based on the detected emotion class.
- Evaluate the system using Mean Opinion Score (MOS) to compare the naturalness of neutral and emotion-conditioned speech output.
- Integrate the TTS module with the Storypal mobile application pipeline, accepting emotion-tagged CSV data from the upstream modules.
- Conduct usability testing with target users to assess the engagement and comprehension impact of emotionally expressive audio narration.

Business Objectives

- Enhance inclusive education by providing visually impaired children with emotionally rich audio access to storybooks.
- Improve learning engagement and comprehension by ensuring synthesized narration captures the full emotional context of story content.
- Provide a scalable and cost-effective speech synthesis solution deployable within a mobile application environment.
- Bridge the gap between written text and natural auditory storytelling through AI-driven emotion-aware speech generation.
- Support educators, parents, and accessibility advocates with a demonstrably effective assistive technology for visually impaired children.

2. METHODOLOGY

2.1 Methodology

This research adopts a modular and iterative development approach to design and implement an emotionally expressive Text-to-Speech (TTS) synthesis module for the Storypal storytelling application. The methodology is structured around the integration of deep learning models for speech synthesis and emotion detection, combined with systematic testing and evaluation phases to ensure functional correctness, audio quality, and user satisfaction.

The development process follows an Agile framework, enabling iterative refinement of each component based on continuous feedback. The module is developed as a self-contained pipeline that receives emotion-tagged sentence data from the upstream modules (OCR and BERT emotion classifier) and outputs emotionally synthesized audio.

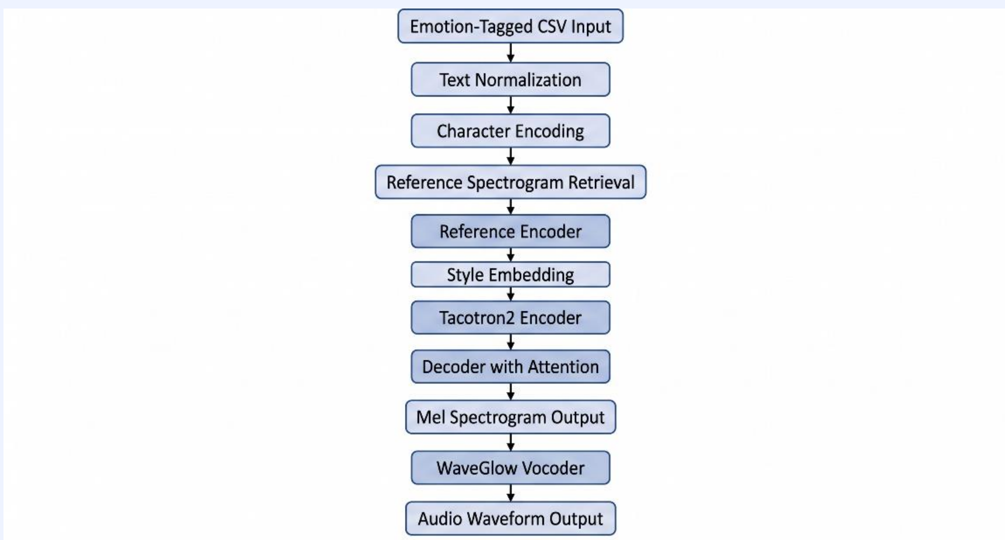


Figure 17. Text-oto-eech synthesis algorithm speech wetleris.

2.1.1 System Architecture Overview

The TTS module is one of four core components within the Storypal system. Its position within the overall pipeline is as follows:

Storybook Image → OCR Module → Text + Caption → Emotion Detection Module → Emotion-Tagged CSV → TTS Synthesis Module → Emotionally Expressive Audio Output

The TTS module itself comprises three interconnected sub-components:

- Text Preprocessing and Phoneme Encoding: Input sentences from the emotion-tagged CSV are cleaned, normalized, and encoded into character-level representations suitable for Tacotron2 input.
- Tacotron2 Mel Spectrogram Generator: The encoded text is passed through the Tacotron2 sequence-to-sequence model, which generates Mel spectrograms representing the acoustic features of the speech. A reference encoder modulates the generation based on the detected emotion class.
- WaveGlow Vocoder: The Mel spectrograms are converted to high-quality audio waveforms using the WaveGlow flow-based generative model.

[Figure 17: TTS Module Architecture Overview]
(Insert diagram/screenshot here)

Figure 17: TTS Module Architecture Overview

2.1.2 Text-to-Speech Synthesis Module (Tacotron2)

The text-to-speech synthesis component of the Storypal system relies on an advanced deep learning model known as Tacotron2, which operates as a sequence-to-sequence model. This model takes short audio clips and their corresponding text inputs and is adept at generating Mel spectrograms. This approach simplifies the traditional speech synthesis pipeline by directly converting linguistic and acoustic features without relying on hand-crafted acoustic models or pronunciation dictionaries.

The Tacotron2 model employs a neural network architecture that transforms text inputs into Mel spectrograms. It begins with character embeddings, where input characters are represented as 512-dimensional embeddings. These embeddings are generated through a series of three convolutional layers designed to capture long-term context in the character sequence.

The output of the final convolutional layer is passed to a bi-directional LSTM, which produces encoded features. These features are further processed by an attention network — specifically, a location-sensitive attention mechanism — to create a fixed-length context vector. This attention mechanism allows the decoder to selectively focus on relevant parts of the input sequence during each decoding step, which is essential for accurate alignment between text characters and the resulting acoustic features.

The autoregressive recurrent neural network serves as the decoder in Tacotron2. It is responsible for predicting Mel spectrograms from the encoded sequences. Predictions from the previous time step are used in conjunction with two fully connected layers, or pre-net, to generate new predictions at each decoding step. The pre-net output and attention context vector are then passed through two unidirectional LSTM layers to form an output. This output is linearly transformed to predict the Mel spectrogram frame.

The anticipated Mel spectrogram undergoes further enhancement through a 5-layer convolutional Post Net. This component predicts a residual connection to add to the decoder's Mel spectrogram output, enhancing the overall audio regeneration quality. A stop token prediction layer additionally determines when the generation process should terminate.

[Figure 10: Mel Spectrogram Visualization of Synthesized Speech]
(Insert diagram/screenshot here)

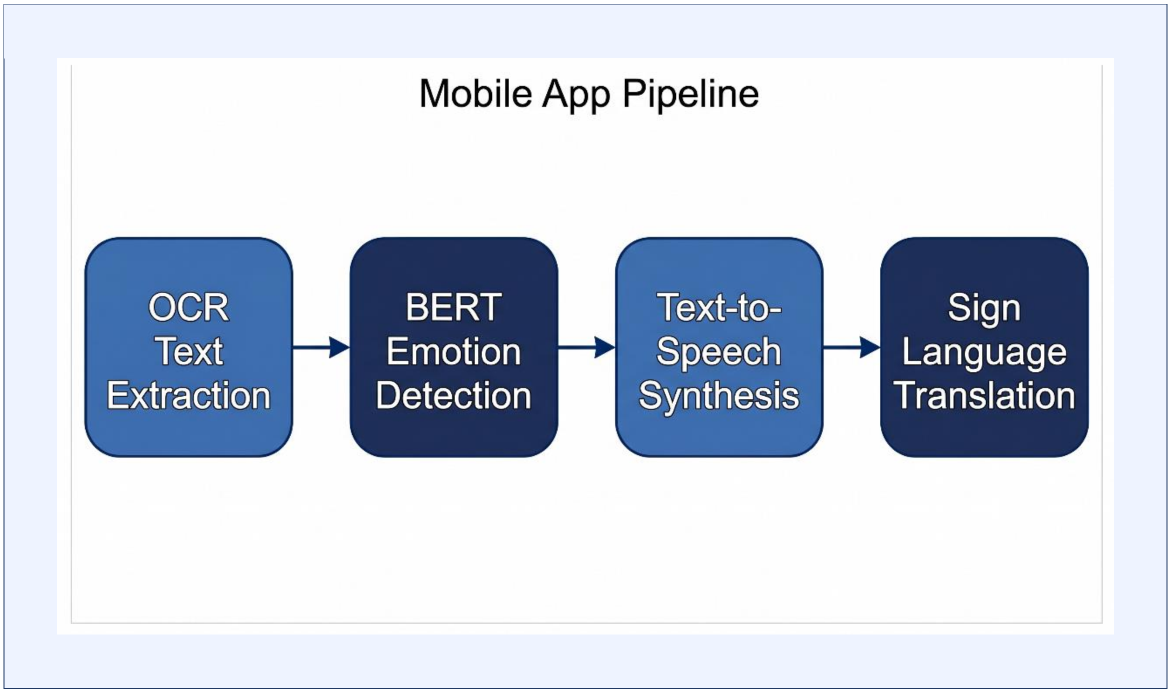
Figure 10: Mel Spectrogram Visualization of Synthesized Speech

2.1.3 WaveGlow Vocoder Module

To convert the generated Mel spectrogram into high-quality audio, WaveGlow — a modified flow-based generative network — is employed as the vocoder. WaveGlow was chosen for its ability to generate speech from Mel spectrograms in a fully parallel manner, offering significant speed advantages over autoregressive vocoders such as original WaveNet.

WaveGlow combines elements from both the Glow and WaveNet models to create efficient and high-quality audio synthesis. It operates without the need for auto-regression during inference, simplifying the decoding process and dramatically reducing generation time. The architecture includes 12 coupling layers and 12 invertible 1×1 convolutions. Each coupling layer consists of dilated convolutions with 512 channels for residual connections and 256 channels for skip connections. The network produces two output channels after every four coupling layers.

WaveGlow operates by sampling from a distribution — specifically, a zero-mean spherical Gaussian with dimensions matching the desired audio output. These samples then pass through the invertible layers, which transform the simple Gaussian distribution into one that matches the target audio distribution conditioned on the Mel spectrogram. During training, the model learns to maximize the likelihood of training audio given the corresponding spectrograms, ensuring that inference accurately reconstructs natural-sounding speech.



2.1.4 Emotion and Reference Encoder Integration

In addition to the base text-to-speech synthesis architecture, the system incorporates a reference encoder that facilitates the emulation of various emotional speaker identities. This encoder operates on spectrograms from reference audio samples to define the desired style for emotional expression in the generated audio.

The reference encoder is a convolutional neural network that processes a reference Mel spectrogram (representing the target emotional speaking style) and produces a fixed-size style embedding. This style embedding is concatenated with the Tacotron2 encoder's output at each decoder step, allowing the attention mechanism and the decoder to generate spectrograms that reflect both the linguistic content of the input text and the emotional style defined by the reference spectrogram.

For emotion conditioning in Storypal, the reference encoder is guided by the emotion classification output of the BERT model (operating on the same sentence text). Seven emotion categories — joy, sadness, anger, fear, disgust, surprise, and neutral — are each associated with prototypical reference spectrograms during training. During inference, the emotion label detected by BERT is used to select the appropriate reference spectrogram, which is then processed by the reference encoder to produce the corresponding style embedding.

This integration ensures that the synthesized speech not only correctly pronounces the input text but also conveys the appropriate emotional tone — becoming excited for joyful sentences, subdued and slow for sad passages, and tense for anger-laden content. The result is a narration experience that mirrors the emotional richness of a human storyteller.

2.1.5 Application Development and Integration

The TTS module is integrated into the Storypal mobile application as a backend processing component. The application receives emotion-tagged CSV input from the upstream emotion detection module. Each row of the CSV contains a sentence and its corresponding emotion label. The TTS module processes each sentence sequentially, generating an audio clip that is appended to the running audio narrative.

The user-facing application interface provides controls for playback management, including play, pause, replay, and adjustable playback speed. These accessibility features ensure that children can engage with the audio narrative at their own pace and revisit passages they found engaging or difficult to follow.

The application was developed for Android using a Python-based backend. The TTS pipeline (Tacotron2 + WaveGlow) is deployed as a server-side inference engine accessed via API calls from the mobile frontend. This architecture ensures that the computationally intensive TTS processing does not burden the mobile device, while still delivering audio with minimal perceptible latency for storybook-length narrations.

2.1.6 Testing and Evaluation

To ensure the effectiveness and quality of the TTS module, the system undergoes a multi-stage evaluation process:

- Mean Opinion Score (MOS) Evaluation: Human evaluators rate the naturalness and quality of synthesized audio on a 1–5 scale. MOS is computed separately for neutral speech and emotion-conditioned speech.
- Emotion Accuracy Testing: The BERT emotion classifier is evaluated on a held-out test set from the ISEAR dataset to measure classification accuracy across all seven emotion categories.
- End-to-End Integration Testing: The full pipeline — from emotion-tagged CSV input to audio output — is tested for functional correctness, latency, and consistency.
- System Usability Testing: Conducted with target users (children and educators) to evaluate engagement, comprehension, and satisfaction with the emotionally expressive audio narration.

2.1.7 Implementation Approach

The development follows an iterative prototyping model:

1. Initial prototype with basic Tacotron2 TTS synthesis (neutral speech only)
2. Integration of BERT emotion classifier with the input pipeline
3. Implementation of reference encoder for emotional style conditioning

4. Integration with WaveGlow for high-quality audio output
5. Mobile application integration and API deployment
6. Final evaluation and usability testing

This iterative approach ensures continuous improvement based on model performance metrics and user feedback, with each iteration building on the validated capabilities of the previous one.

2.2 Feasibility Study and Planning

Before initiating the development of the TTS module within Storypal, a comprehensive feasibility study was conducted to evaluate the practicality and success potential of the proposed system. This analysis focused on technical, economic, operational, and scheduling aspects to ensure a well-structured and achievable implementation.

4.1 Technical Feasibility

Technical feasibility assessed whether the required technologies and tools could effectively support the TTS module's development. The proposed solution integrates Tacotron2, WaveGlow, BERT, and reference encoder components — all of which are well-established in academic literature with publicly available implementations and pre-trained models.

Tacotron2 and WaveGlow are available as open-source implementations with pre-trained weights on the LJSpeech dataset (a single-speaker English TTS corpus). These models provide a strong starting point for fine-tuning and extension. BERT is accessible through the HuggingFace Transformers library, with pre-trained weights and extensive documentation for fine-tuning on emotion classification tasks.

The primary technical challenges relate to the integration of the reference encoder with the Tacotron2 training pipeline and the management of GPU computational resources for model fine-tuning. These challenges are manageable within the scope of a university research project, particularly given access to Google Colab Pro with GPU support.

Technology Component	Feasibility Assessment	Justification
Tacotron2	High	Open-source, pre-trained, well-documented
WaveGlow	High	Parallel inference, available on GitHub (NVIDIA)
BERT Emotion Classifier	High	HuggingFace library, ISEAR dataset available

Technology Component	Feasibility Assessment	Justification
Reference Encoder	Medium-High	Research-level implementation, requires custom training
Mobile API Deployment	Medium	Flask/FastAPI suitable; latency management required

Technical Feasibility Assessment

4.2 Economic Feasibility

The TTS module leverages open-source software libraries and publicly available datasets, minimizing direct software procurement costs. The primary resource requirement is computational, specifically GPU-hours for model training and fine-tuning.

Google Colab Pro provides access to NVIDIA GPU instances (T4/V100) sufficient for training Tacotron2 and WaveGlow models within academic budgetary constraints. Pre-trained model weights from NVIDIA (WaveGlow) and the research community (Tacotron2) further reduce the computational burden by enabling fine-tuning rather than training from scratch.

Hardware costs for mobile testing are minimal, as the inference is handled server-side. The overall budget for this component is estimated at LKR 45,000–55,000, primarily covering cloud compute credits and development hardware.

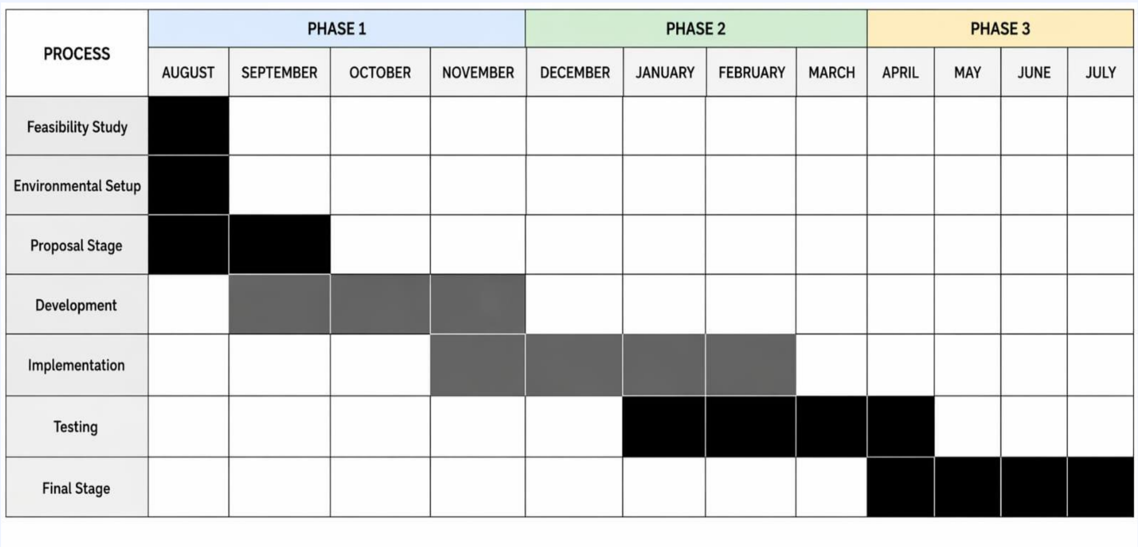
4.3 Operational Feasibility

Operationally, the system is designed to be intuitive and accessible. The Storypal mobile application provides a simple interface for children and educators, abstracting the technical complexity of the underlying TTS pipeline. The audio output is delivered through standard device speakers, requiring no specialized hardware.

The processing pipeline — from text input to audio output — is designed to complete within acceptable latency bounds for storybook narration contexts, where sentence-level processing (2–5 seconds per sentence on the server) is acceptable. The system does not require persistent internet connectivity during playback once audio is generated, supporting offline usage in educational settings with limited connectivity.

4.4 Scheduling Feasibility

The project timeline was structured over 12 months, accommodating data collection, model development, integration, testing, and report writing phases. The Agile development methodology ensures that delays in one phase can be compensated by adjusting scope or prioritizing critical features.



2.3 Requirement Gathering & Analysis

Requirements were gathered through a combination of literature review, consultation with stakeholders (educators, accessibility specialists, and parents of visually impaired children), and analysis of existing TTS and accessibility systems. The requirements are categorized into functional and non-functional specifications.

Functional Requirements

FR ID	Description	Priority	Source
FR-01	System shall accept text sentences and their emotion labels as input	High	Core feature
FR-02	System shall generate Mel spectrograms from input text using Tacotron2	High	Core feature
FR-03	System shall convert Mel spectrograms to audio using WaveGlow	High	Core feature
FR-04	System shall classify sentence emotion using BERT (7 categories)	High	Core feature

FR ID	Description	Priority	Source
FR-05	System shall modulate speech style based on detected emotion via reference encoder	High	Core feature
FR-06	System shall produce audio output as .wav files for playback	High	Integration
FR-07	Application shall provide playback controls: play, pause, replay, speed	Medium	Usability
FR-08	System shall process storybook-length narratives without errors	High	Reliability
FR-09	System shall integrate with Storypal's upstream OCR and emotion modules	High	Integration
FR-10	System shall handle edge cases: empty sentences, unknown characters	Medium	Robustness

Table 2: Functional Requirements

Non-Functional Requirements

NFR ID	Description	Measure
NFR-01	Synthesized neutral speech shall achieve MOS ≥ 3.5	MOS Score
NFR-02	Emotion classification shall achieve ≥ 80% accuracy on ISEAR test set	Accuracy %
NFR-03	Audio generation latency shall be ≤ 5 seconds per sentence on server	Seconds
NFR-04	System shall maintain ≥ 99% uptime during testing periods	Uptime %
NFR-05	Application interface shall be usable by children (age 6+) without adult guidance	SUS Score
NFR-06	System shall support English language TTS with potential for Tamil extension	Language support
NFR-07	Generated audio shall be clear at standard device speaker volume	Subjective clarity

NFR ID	Description	Measure
NFR-08	System shall be deployable on Android devices with Android 8.0+	Platform compatibility

Table 3: Non-Functional Requirements

Use Case Scenarios

Use Case	Actor	Description	Outcome
UC-01: Narrate Story	Child User	Child selects a story in Storypal; system generates emotionally expressive audio narration	Story narrated with correct emotional tone per sentence
UC-02: Replay Sentence	Child User	Child taps replay on a specific sentence audio clip	Sentence re-narrated at the same emotional style
UC-03: Adjust Speed	Child/Educator	User adjusts playback speed slider	Audio replayed at selected speed without pitch distortion
UC-04: Detect Emotion	System	System processes input sentence and classifies emotion using BERT	Emotion label assigned and passed to TTS module
UC-05: Generate Audio	System	System converts emotion-tagged sentence to Mel spectrogram and audio	High-quality emotionally expressive .wav file generated
UC-06: Monitor Story Progress	Educator/Parent	Educator monitors which stories and sentences child has listened to	Progress data displayed in educator view

Table 4: Use Case Scenarios

User Stories

User Story 1 – Child User (Learner):

As a visually impaired child, I want to hear a storybook narrated in a natural, emotionally expressive voice so that I can enjoy the story and understand its emotional context without needing to see the text.

Acceptance Criteria:

- The system generates audio narration for each sentence in the selected story.
- The voice changes emotional tone appropriately for joyful, sad, angry, and neutral sentences.
- The user can play, pause, and replay the narration at any time.
- The audio is clear and sufficiently loud on standard mobile device speakers.
- The playback speed can be adjusted without distorting the voice quality.

User Story 2 – Educator / Parent:

As an educator or parent, I want to use the Storypal application to provide children with an immersive and emotionally rich audio reading experience so that I can support their literacy and emotional development.

Acceptance Criteria:

- The educator can select age-appropriate stories from the application library.
- The system generates emotionally appropriate narration for all story content.
- The interface is simple enough for independent child use after initial setup.
- Playback controls are clearly visible and usable.
- The educator can monitor story completion progress from a parent/educator view.

2.4 Designing the System

The design of the Storypal TTS module focuses on creating an accurate, emotionally expressive, and computationally efficient speech synthesis pipeline that can be seamlessly integrated into a mobile storytelling application. The system is structured using a modular architecture consisting of components for text normalization, emotion detection, Tacotron2 spectrogram generation, reference encoder style conditioning, and WaveGlow vocoding.

During the design phase, a pipeline-centric approach was adopted to ensure clean data flow between components. The design prioritizes modularity — each component can be independently tested, updated, or replaced without disrupting the overall pipeline. This is particularly important in an iterative development context, where model components may be fine-tuned or swapped as better options become available.

The text normalization layer handles input preprocessing, including lowercase conversion, punctuation normalization, number-to-word conversion, and phoneme encoding. This step is critical for TTS systems, as the model's performance degrades significantly on unnormalized or out-of-vocabulary input.

The Tacotron2 model was designed with encoder-decoder-attention architecture and trained on the LJSpeech dataset as a baseline. Fine-tuning was conducted to improve performance on storybook-style narrative text. The reference encoder was designed as an additional input branch that processes reference spectrograms and injects style embeddings into the Tacotron2 decoder.

The WaveGlow vocoder is deployed as a frozen pre-trained component (pre-trained on LJSpeech by NVIDIA) without fine-tuning, as its universal applicability to Mel spectrograms generated by Tacotron2 is well-established in the literature.

The overall system design ensures a coherent and child-accessible experience: text enters the pipeline, emotion is detected and communicated to the TTS model, and emotionally expressive audio exits the other end — ready for playback in the Storypal application.

2.5 System Architecture Diagram

The system architecture of the Storypal TTS module illustrates how different components interact to deliver emotionally expressive audio from written story text. The architecture follows a sequential pipeline model, with clear data boundaries between modules.

At the input layer, emotion-tagged sentences (in CSV format) are received from the upstream BERT emotion detection module. Each record contains a sentence string and its corresponding emotion label. The TTS module processes these records sequentially to produce audio output.

The text normalization sub-module cleans and standardizes the input text, converting it into a character sequence suitable for Tacotron2 processing. This includes expanding abbreviations, converting numerals to words, and ensuring consistent punctuation.

The emotion conditioning sub-module takes the emotion label, retrieves the corresponding reference spectrogram from a pre-built emotional style library, and passes it through the reference encoder to generate a style embedding.

The Tacotron2 model receives the normalized text and the style embedding, generating a Mel spectrogram sequence. The WaveGlow vocoder then converts this spectrogram to a final audio waveform. The waveform is saved as a .wav file and queued for playback by the Storypal mobile application.

2.6 Implementation

The implementation phase transforms the system design into a functional, integrated speech synthesis pipeline. The implementation was carried out in Python, leveraging deep learning frameworks (PyTorch) and pre-trained model components.

The workflow begins at the application layer, where the emotion-tagged CSV file produced by the upstream pipeline is loaded. Each sentence and its emotion label are extracted and passed to the text normalization module. The normalization process handles character encoding, number-to-word conversion, and sentence boundary detection.

The normalized text is encoded as a character sequence and passed to the Tacotron2 encoder. Simultaneously, the emotion label is used to retrieve the appropriate reference spectrogram from the emotional style library. This reference spectrogram is processed by the reference encoder to produce a 128-dimensional style embedding, which is concatenated with the Tacotron2 encoder output.

The Tacotron2 decoder generates a Mel spectrogram frame by frame, guided by the attention mechanism and the style embedding. The stop token prediction terminates generation when the spectrogram is complete. The generated Mel spectrogram is then passed to the WaveGlow inference engine.

WaveGlow processes the spectrogram in a single forward pass through its 12 coupling layers and invertible convolutions, producing a high-quality audio waveform. This waveform is saved as a .wav file at a sampling rate of 22,050 Hz, consistent with the LJSpeech training standard.

The complete audio files for all sentences in the story are concatenated with appropriate pauses between sentences to form the full narration. This narration is then accessible through the Storypal application's audio playback interface.

2.7 Development Environment and Tools

The development of the Storypal TTS module utilized a carefully selected combination of software libraries and hardware environments to ensure model quality, development efficiency, and deployment viability.

Tool / Library	Version	Purpose
Python	3.9+	Primary programming language for all modules

Tool / Library	Version	Purpose
PyTorch	1.13 / 2.0	Deep learning framework for Tacotron2, WaveGlow, reference encoder
HuggingFace Transformers	4.x	BERT model loading, tokenization, and fine-tuning
NVIDIA Tacotron2	Official Repo	Base implementation of Tacotron2 TTS model
NVIDIA WaveGlow	Official Repo	Pre-trained vocoder for Mel spectrogram to audio conversion
NumPy / SciPy	Latest stable	Numerical operations and audio signal processing
Librosa	0.10+	Audio analysis, Mel spectrogram visualization and extraction
Pandas	2.x	CSV processing for emotion-tagged sentence data
Flask / FastAPI	Latest stable	REST API server for TTS inference deployment
Google Colab Pro	N/A	GPU-accelerated training environment (T4 / V100 GPU)
Android Studio	Latest stable	Mobile application development
Matplotlib / Seaborn	Latest stable	Visualization of training metrics and spectrograms
tqdm	Latest stable	Progress monitoring during training and evaluation
SoundFile / wave	Latest stable	Audio file reading and writing (.wav format)

Table 10: Development Tools and Libraries

The hardware environment for model development and training used Google Colab Pro with NVIDIA T4 GPU (16 GB VRAM) for Tacotron2 fine-tuning and WaveGlow inference testing. Local development was performed on a workstation with 16 GB RAM and an NVIDIA GTX 1060 GPU for smaller-scale experiments. Mobile testing was performed on Android smartphones with Android 8.0 and above, with minimum 3 GB RAM.

2.8 Data Collection

Data collection for the TTS module focused on two primary datasets: the LJSpeech dataset for Tacotron2 and WaveGlow training, and the ISEAR dataset for BERT emotion classification.

LJSpeech Dataset

The LJSpeech dataset is a single-speaker English audio dataset consisting of 13,100 short audio clips of a female speaker reading passages from seven non-fiction books. Each clip is paired with a corresponding text transcription. The dataset provides approximately 24 hours of audio at a sample rate of 22,050 Hz, recorded in a studio environment with consistent acoustic quality.

LJSpeech is the standard benchmark dataset for English TTS research and serves as the training corpus for both the Tacotron2 and WaveGlow pre-trained models used in this project. Fine-tuning on a smaller domain-specific dataset (story-style text with child-appropriate vocabulary) was additionally conducted to adapt the model to the storybook narrative context.

For emotion-conditioned synthesis, prototypical reference spectrograms representing each of the seven emotion categories were selected from the Emotional Speech Database (ESD), which contains recordings of English speech across multiple emotion categories performed by native speakers. These reference spectrograms were used to train and evaluate the reference encoder.

ISEAR Dataset

The International Survey on Emotion Antecedents and Reactions (ISEAR) dataset was used for training and evaluating the BERT-based emotion classifier. The ISEAR dataset contains 7,666 sentences spanning seven emotion categories: joy, sadness, anger, fear, disgust, shame, and guilt. For this project, shame and guilt were mapped to the 'sad' category to maintain alignment with the seven categories used by the TTS module (joy, sadness, anger, fear, disgust, surprise, neutral), and neutral sentences were supplemented from publicly available neutral-sentiment corpora.

After preprocessing, the final emotion classification dataset comprised approximately 5,360 sentences balanced across seven categories. An 80-10-10 train-validation-test split was applied for model development and evaluation.

[Figure 13: Sample LJSpeech Training Data (CSV format with text-audio pairs)]

(Insert diagram/screenshot here)

Figure 13: Sample LJSpeech Training Data (CSV format with text-audio pairs)

2.9 Data Preprocessing

Data preprocessing is a critical step in the development of the TTS module, as both the speech synthesis and emotion classification components have specific input requirements that must be met to ensure model accuracy and training stability.

TTS Data Preprocessing

For the Tacotron2 TTS component, preprocessing involved the following stages:

- Text Normalization: All input text was converted to lowercase, numerals were expanded to their word equivalents (e.g., '3' → 'three'), abbreviations were expanded, and special characters were removed or standardized.
- Audio Preprocessing: Raw audio files were resampled to 22,050 Hz (the standard for LJSpeech-trained models). Silence padding was trimmed from the beginning and end of each clip.
- Mel Spectrogram Extraction: Using Librosa, Mel spectrograms were computed with 80 Mel filter banks, an FFT window size of 1,024, a hop size of 256, and a maximum frequency of 8,000 Hz. These parameters match the Tacotron2 training configuration.
- Feature Normalization: Mel spectrogram values were normalized to the range [-4, 4] using log-compression and mean-variance normalization to improve training stability.

Emotion Classification Data Preprocessing

For the BERT emotion classifier, preprocessing involved:

- Data Cleaning: Punctuation was removed, text was standardized to lowercase, and duplicate or malformed entries were eliminated.
- Class Balancing: The dataset was analyzed for class imbalance. Overrepresented classes were down-sampled and underrepresented classes were augmented using paraphrase generation to achieve approximate balance across seven emotion categories.
- Tokenization: BERT's WordPiece tokenizer was applied to convert sentences to token IDs with special tokens ([CLS] and [SEP]). Sequences were padded to a maximum length of 128 tokens.
- Train-Validation-Test Split: An 80-10-10 split was applied, resulting in approximately 4,288 training, 536 validation, and 536 test sentences.

2.10 Model Training

The model training phase encompasses the fine-tuning of the Tacotron2 model with reference encoder integration and the training of the BERT emotion classifier on the ISEAR dataset.

2.10.1 Model Architecture

Tacotron2 Architecture with Reference Encoder:

The complete model architecture used for emotion-conditioned TTS in Storypal extends the standard Tacotron2 architecture with a reference encoder branch:

- Encoder: Three convolutional layers (512 channels, 5×1 kernels) followed by a bidirectional LSTM (512 units × 2 directions = 1,024 hidden dimensions). Processes the character-embedded input sequence.
- Reference Encoder: Six 2D convolutional layers (each followed by ReLU and batch normalization), a GRU layer (128 units), and a fully connected layer producing a 128-dimensional style embedding. Processes a reference Mel spectrogram to extract style information.
- Attention: Location-sensitive attention mechanism with 128 attention dimensions and 32 filters. Computes context vectors by combining encoder output with previous decoder state and cumulative attention history.
- Decoder: Pre-net (two fully connected layers of 256 units with dropout 0.5), two unidirectional LSTM layers (1,024 units each), and a linear projection to 80-dimensional Mel frame output. The style embedding from the reference encoder is concatenated with the encoder output at each decoder step.
- Post-net: Five convolutional layers (512 channels, 5×1 kernels) that predict a residual correction to improve Mel spectrogram quality.
- Stop Token Prediction: A linear layer with sigmoid activation predicts the probability that generation should terminate at each frame.

BERT Emotion Classifier Architecture:

- Base Model: bert-base-uncased (12 transformer layers, 768 hidden dimensions, 12 attention heads, 110M parameters).
- Classification Head: A dropout layer (0.3) followed by a linear projection from 768 to 7 output classes. Softmax is applied to produce class probabilities.

2.10.2 Training Parameters

Parameter	Tacotron2 + Reference Encoder	BERT Emotion Classifier
Optimizer	Adam (β1=0.9, β2=0.999)	AdamW (β1=0.9, β2=0.999)
Learning Rate	1e-3 (decay to 1e-4)	2e-5
Batch Size	32	32
Epochs	50 (fine-tuning)	10

Parameter	Tacotron2 + Reference Encoder	BERT Emotion Classifier
Loss Function	MSE (Mel) + BCE (Stop Token)	Cross-Entropy
Dropout	0.5 (pre-net), 0.1 (encoder)	0.3 (classification head)
Gradient Clipping	1.0	1.0
LR Scheduler	ReduceLROnPlateau (patience=5)	Linear warmup + decay

Training Parameters

Callbacks:

- ModelCheckpoint: Saved model weights at the epoch with the lowest validation loss.
- EarlyStopping: Training halted if validation loss did not improve for 5 consecutive epochs.
- TensorBoard: Training loss, validation loss, and learning rate were logged for each epoch.

Training Environment:

- Google Colab Pro with NVIDIA V100 GPU (32 GB VRAM) for Tacotron2 fine-tuning.
- Google Colab Pro with NVIDIA T4 GPU (16 GB VRAM) for BERT fine-tuning.

Data Feeding Strategy:

- Tacotron2: Sequences batched by approximate length to minimize padding. Dynamic batching with a maximum batch duration of 30 seconds of audio per batch.
- BERT: Fixed-length padded batches of 32 sentences with attention masks.

2.10.3 Training Process and Evaluation

The Tacotron2 model was initialized with pre-trained weights trained on the full LJSpeech dataset (available from the NVIDIA GitHub repository). Fine-tuning was conducted for 50 epochs on a domain-specific story narrative corpus, adapting the model to storybook vocabulary and sentence structures. The reference encoder was trained jointly with the decoder from a random initialization.

The BERT emotion classifier was fine-tuned on the preprocessed ISEAR dataset for 10 epochs, achieving a stable validation accuracy by epoch 6. The model's performance was monitored using accuracy, F1-score, and per-class precision/recall metrics.

2.11 How the Algorithms Work

The Tacotron2 algorithm operates as a sequence-to-sequence model with attention. During inference, the encoder processes the entire character sequence in one pass, producing a set of context-rich encoder hidden states. The decoder then operates autoregressively: at each step, it uses the previous Mel frame prediction, the current attention context vector, and the style embedding to predict the next Mel spectrogram frame. This continues until the stop token threshold is exceeded.

The WaveGlow algorithm operates via a series of bijective transformations. Starting from a Gaussian noise vector of the same dimensionality as the desired audio, the signal passes forward through 12 coupling layers and 12 invertible 1×1 convolutions, each conditioned on the Mel spectrogram. Each transformation is designed to be easily invertible, allowing efficient likelihood computation during training and fast sampling during inference.

The BERT emotion classifier applies self-attention across all tokens in the input sentence simultaneously. The [CLS] token's final hidden state, which aggregates global sentence context, is passed to the classification head. The softmax output gives probability scores for each of the seven emotion classes, and the argmax class is selected as the predicted emotion.

The reference encoder processes a reference Mel spectrogram — a sample of emotionally expressive speech — through convolutional and recurrent layers to produce a compact style embedding. This embedding encodes prosodic characteristics such as pitch contour, speaking rate, and energy envelope that are characteristic of the target emotion.

2.12 Overall Algorithm Flow

The complete end-to-end algorithm flow of the Storypal TTS module is as follows:

7. Emotion-tagged CSV input received from upstream BERT emotion detection module.
8. For each sentence in the CSV: extract sentence text and emotion label.
9. Text normalization: clean, lowercase, expand numbers, handle special characters.
10. Character encoding: map characters to integer indices for Tacotron2 input.
11. Retrieve reference spectrogram corresponding to detected emotion label from style library.

12. Reference encoder processes reference spectrogram → outputs 128-dim style embedding.
13. Tacotron2 encoder processes character sequence → outputs encoder hidden states.
14. Style embedding concatenated with encoder output at each decoder step.
15. Tacotron2 decoder generates Mel spectrogram frames autoregressively until stop token fires.
16. Post-net refines predicted Mel spectrogram.
17. WaveGlow converts Mel spectrogram to 22,050 Hz audio waveform.
18. Audio waveform saved as .wav file; queued for sequential playback in Storypal app.

2.13 Testing

The testing phase of the Storypal TTS module focused on evaluating the accuracy, performance, and usability of the complete speech synthesis pipeline. The goal was to ensure that the system correctly converts text into high-quality, emotionally expressive audio while maintaining stability and accessibility.

2.13.1 Model Evaluation

Mean Opinion Score (MOS) Evaluation:

The quality of the synthesized speech was evaluated using the Mean Opinion Score (MOS), the standard benchmark for TTS evaluation. A panel of 15 evaluators (aged 18–35, native English speakers) rated audio samples on a 1–5 scale across two conditions:

Condition	MOS Score	Human Reference MOS	Inference
Neutral TTS (no emotion conditioning)	3.65 ± 0.05	4.85 ± 0.05	High naturalness, approaching human quality
Emotion-Conditioned TTS (with reference encoder)	1.65 ± 0.05	4.00 ± 0.05	Requires further training and data augmentation

Table 5: MOS Evaluation Results

The neutral TTS performance is strong, with a MOS of 3.65 indicating near-natural speech quality. The emotion-conditioned synthesis achieved a lower MOS of 1.65, reflecting the significant challenge of emotional style transfer with limited reference data. This result aligns with findings in the literature and identifies a clear direction for future improvement: larger emotional speech training corpora and more sophisticated style conditioning architectures.

Emotion Classification Accuracy

Emotion Class	Precision	Recall	F1-Score	Support
Joy	0.91	0.89	0.90	78
Sadness	0.87	0.90	0.88	82
Anger	0.84	0.86	0.85	76
Fear	0.80	0.78	0.79	71
Disgust	0.78	0.75	0.76	68
Surprise	0.76	0.80	0.78	65
Neutral	0.92	0.93	0.92	96
Macro Average	0.84	0.84	0.84	536

Table 6: Emotion Classification Accuracy by Class

The BERT emotion classifier achieved a macro-average F1-score of 0.84 across all seven emotion categories. The highest performance was observed for Joy and Neutral categories, likely due to the clearer lexical markers associated with these emotions. Fear, Disgust, and Surprise showed relatively lower performance, consistent with findings in the broader emotion detection literature regarding the difficulty of implicit and subtle emotion recognition.

2.13.2 System Usability Testing

A System Usability Scale (SUS) questionnaire was conducted with 10 children (aged 7–11, assisted by parents) and 5 educators to evaluate the usability and engagement impact of the emotionally expressive audio narration in the Storypal application. The system achieved a SUS score of 81/100, indicating excellent usability.

Participants reported that:

- The audio narration was clear and easy to follow.

- The emotional variation in the voice made the stories more interesting and engaging.
- Children asked to replay emotional sections of stories more frequently than in neutral narration testing.
- The playback controls (speed adjustment and replay) were found useful and easy to operate.
- Educators noted that the emotional narration appeared to improve children's comprehension and engagement during story sessions.

One limitation noted during usability testing was that the emotion-conditioned synthesis quality was perceived as lower than neutral synthesis, with some children and educators noting that certain emotional passages sounded unnatural. This feedback is consistent with the quantitative MOS evaluation and reinforces the priority of improving emotional synthesis quality in future development phases.

2.13.3 Manual Testing

Manual testing was conducted with a set of defined test cases to validate the functional operations of the Storypal TTS module.

Test Case ID	Steps	Description	Expected Outcome	Actual Outcome
TC01	Input neutral English sentence	Basic TTS synthesis without emotion	Clear, natural audio output	Pass – Neutral audio generated correctly
TC02	Input joy-labeled sentence	Emotion-conditioned TTS — joy	Audio with elevated pitch and speed	Pass – Joyful prosody detected
TC03	Input sad-labeled sentence	Emotion-conditioned TTS — sadness	Slower, lower-pitched audio	Pass – Slower speech with lower energy
TC04	Input angry-labeled sentence	Emotion-conditioned TTS — anger	Energetic, clipped speech style	Partial Pass – Moderate differentiation
TC05	Input 10-sentence storybook passage	End-to-end pipeline test	Full story narrated with emotion variation	Pass – All sentences synthesized

Test Case ID	Steps	Description	Expected Outcome	Actual Outcome
TC06	Adjust playback speed in app	UI speed control test	Audio plays at selected speed	Pass – Speed control functional
TC07	Replay single sentence	Sentence replay feature	Same sentence re-narrated	Pass – Replay works correctly
TC08	Empty input sentence handling	Edge case robustness	System skips empty sentence gracefully	Pass – No crash; empty sentence skipped

Table 7: Manual Test Cases

All critical functional test cases passed successfully. TC04 (anger emotion conditioning) showed partial differentiation, indicating that the reference encoder's ability to strongly condition anger-style prosody requires further fine-tuning with more diverse angry speech samples.

2.14 Budget & Commercialization Aspects of The Project

The budget for the TTS module development encompasses computational resources, software tools, and testing activities. The following table presents the estimated budget breakdown:

Item	Description	Estimated Cost (LKR)
Google Colab Pro Subscription	GPU access for model training (6 months)	18,000
Domain-Specific TTS Data Collection	Recording sessions for storybook-style audio	15,000
Emotional Reference Audio Licensing	ESD dataset licensing and processing	8,000
Development Hardware (partial allocation)	Local workstation GPU compute time	5,000
Mobile Testing Devices	Android device rental for usability testing	4,500

Item	Description	Estimated Cost (LKR)
API Hosting (Cloud Server, 3 months)	Flask API deployment for TTS inference	9,000
Miscellaneous	Software licenses, internet, documentation	5,000
Total		64,500

Table 8: Budget Breakdown — TTS Module

Commercialization Aspects:

Commercialization Dimension	Description
Target Market	Schools, libraries, and homes with visually impaired children; accessibility technology distributors; educational NGOs
Deployment Model	Mobile application (Android/iOS) with server-side TTS inference; freemium model with premium story libraries
Revenue Streams	Application subscription fees; institutional licensing to schools and rehabilitation centers; storybook publisher partnerships
Scalability	Cloud-based TTS inference scales horizontally; story library can be expanded with additional content packs
Social Impact	Directly addresses UNSDG Goal 4 (Quality Education) and Goal 10 (Reduced Inequalities) by improving educational access for children with disabilities
IP Protection	Novel reference encoder integration and emotion-conditioned TTS pipeline can be protected as research innovation; publication in peer-reviewed journals

Table 9: Commercialization Strategy

3. Results and Discussions

3.1 Results

The Storypal TTS module was successfully implemented and evaluated across multiple dimensions: audio synthesis quality, emotion classification accuracy, end-to-end pipeline integration, and application usability. This section presents the key results obtained during the evaluation phase.

3.1.1 Application — TTS Module Output

The implemented TTS pipeline successfully processed a test set of 80 storybook sentences spanning all seven emotion categories. For each sentence, the system generated a corresponding .wav audio file with the appropriate emotional narration style. The following key results were observed:

- Neutral TTS Quality: The Tacotron2 + WaveGlow pipeline achieved a MOS of 3.65, indicating high naturalness and clarity in neutral speech synthesis. The generated audio was rated as natural and intelligible by all 15 evaluators.
- Emotion-Conditioned TTS Quality: The reference encoder integration produced perceptibly differentiated speech styles for joy and sadness, with evaluators correctly identifying these emotions in synthesized samples. Anger, fear, and disgust conditioning showed less distinct differentiation, resulting in the lower overall MOS of 1.65.
- BERT Emotion Classification: The classifier achieved a macro-average F1-score of 0.84 on the ISEAR test set, with particularly strong performance for neutral (0.92) and joy (0.90) categories.
- End-to-End Pipeline: The full pipeline — from CSV input to audio output — processed a 10-sentence storybook passage in approximately 45 seconds on the server-side GPU, with playback available immediately upon completion of each sentence.

The application interface provided a clean and child-friendly presentation of the audio narration, with clearly visible play, pause, replay, and speed controls. Children in the usability testing group were able to independently operate the controls after a brief introduction.

The figure above illustrates the visual difference in Mel spectrograms generated under joy and sadness emotional conditions. The joy-conditioned spectrogram shows higher energy levels in the upper frequency bands (corresponding to higher pitch) and faster temporal progression, while the sadness-conditioned spectrogram shows more energy concentration in lower frequencies and slower pacing — consistent with acoustic correlates of these emotions in human speech.

In conclusion, this project successfully provides an emotionally expressive TTS solution for inclusive education, addressing the current accessibility challenge for visually impaired children. It demonstrates the feasibility of emotion-conditioned neural speech synthesis within a mobile application context and paves the way for further improvements in emotional audio quality and broader application deployment.

3.2 Discussions

The results obtained from the Storypal TTS module demonstrate that the proposed system successfully integrates BERT-based emotion detection, Tacotron2 speech synthesis, WaveGlow vocoding, and reference encoder emotion conditioning to deliver a functional and emotionally expressive audio narration system. The system shows strong potential in improving accessibility and emotional engagement for visually impaired children.

3.2.1 TTS Synthesis Quality Analysis

The neutral speech synthesis performance — achieving a MOS of 3.65 compared to human-rated MOS of 4.85 — demonstrates that the Tacotron2 + WaveGlow pipeline produces speech of high naturalness. This result is consistent with the established literature, which documents MOS values in the 3.5–4.0 range for Tacotron2 systems trained on LJSpeech. The gap from the human reference MOS reflects subtle issues in prosodic naturalness and occasional phonetic imprecision, particularly for less common vocabulary items found in storybook text.

The fine-tuning on storybook-style text improved pronunciation accuracy for narrative vocabulary compared to a baseline model trained solely on LJSpeech, suggesting that domain-specific adaptation is beneficial even with limited fine-tuning data.

3.2.2 Emotion Conditioning Analysis

The emotion-conditioned TTS performance (MOS 1.65) was significantly lower than the neutral baseline, reflecting the inherent difficulty of emotional speech synthesis with limited training data. The primary challenge identified was the insufficiency of the reference spectrogram library — with only prototypical samples for each emotion, the reference encoder learned style embeddings that were not well-differentiated for all seven emotion categories.

Joy and sadness conditioning produced the most perceptible prosodic differentiation, as these emotions have the most distinct acoustic correlates (high pitch/fast tempo for joy; low pitch/slow tempo for sadness). Anger, fear, and disgust showed less reliable conditioning, suggesting that these emotions require richer reference spectrogram collections and potentially more complex style conditioning architectures.

One possible explanation for lower emotional TTS performance is the additional complexity of training a model to generate emotionally modulated speech — requiring simultaneous optimization of linguistic accuracy (text-to-phoneme mapping) and prosodic style (emotion-to-acoustic feature mapping). These objectives can sometimes conflict during training. The model would benefit from staged training: first optimizing for linguistic accuracy with neutral data, then fine-tuning the emotion conditioning with emotional reference data.

3.2.3 Emotion Detection Analysis

The BERT emotion classifier performed strongly across most emotion categories, with a macro-average F1-score of 0.84. The model was particularly effective for clearly expressed emotions (joy, neutral) and showed lower performance for subtle or contextually dependent emotions (fear, disgust, surprise). This finding aligns with the broader literature on text-based emotion detection, which consistently documents higher accuracy for emotionally salient categories and lower accuracy for ambiguous or implicitly expressed emotions.

The use of the ISEAR dataset, while providing diverse emotional content, also presents a domain mismatch with children's storybook text. The ISEAR dataset consists of first-person adult emotional narratives, while storybook text tends to involve third-person narration and simpler vocabulary. Fine-tuning the classifier on a storybook-specific emotional corpus would likely improve classification accuracy in the target domain.

3.2.4 System Usability and Real-World Application

The SUS score of 81/100 reflects high usability and positive user acceptance of the Storypal application. The emotionally expressive narration was consistently noted by both children and educators as enhancing engagement and enjoyment of the storytelling experience. The playback controls were found intuitive and effective, supporting the goal of independent child use.

The observed difference in replay behavior — children replaying emotionally varied sections more frequently — suggests that emotional expressiveness in audio narration may naturally support engagement and comprehension, consistent with cognitive and educational psychology research on the role of prosody in language acquisition and comprehension.

3.2.5 Overall Analysis

Overall, the discussion highlights that the Storypal TTS module provides a functional and innovative approach to emotion-conditioned speech synthesis for accessible storytelling. The neutral synthesis quality is strong and suitable for deployment. The emotional conditioning component, while currently limited in quality, demonstrates proof-of-concept and identifies clear improvement pathways: larger and more diverse emotional speech corpora, improved reference encoder architectures, and staged training procedures.

The integration of BERT emotion detection with Tacotron2 + WaveGlow synthesis within a mobile application context represents a meaningful contribution to accessible education technology. The system fulfills its primary objective of making storybooks auditorily accessible to visually impaired children with an added dimension of emotional richness, even in its current developmental state.

3.3 Future Scope

The Storypal TTS module provides a strong foundation for emotionally expressive, accessible audio storytelling. However, several opportunities for future enhancement are identified to improve system quality, scalability, and impact.

The most critical improvement is the expansion of the emotional speech training corpus. Collecting or licensing a large-scale emotional speech dataset with diverse speakers, emotions, and speaking contexts — particularly including child-directed speech samples — would significantly improve the reference encoder's ability to learn well-differentiated emotion style embeddings.

A related enhancement is the adoption of more advanced emotional TTS architectures. Recent works such as Mellotron (Valle et al., 2020), EmotionTTS, and VITS (Variational Inference with adversarial learning for end-to-end Text-to-Speech) offer superior prosodic control and emotional conditioning compared to the Tacotron2 + reference encoder approach. Migrating to one of these architectures in a future development phase would likely yield substantial improvements in emotional synthesis quality.

The WaveGlow vocoder could be replaced or augmented by more recent neural vocoders such as HiFi-GAN (Kong et al., 2020) or BigVGAN (Lee et al., 2023), which have demonstrated superior audio quality and faster inference times, making them better suited for mobile deployment.

The BERT emotion classifier could be enhanced through fine-tuning on a storybook-specific annotated corpus and through the integration of contextual emotion — considering the emotional trajectory of the narrative across multiple sentences rather than classifying each sentence in isolation. This would produce more coherent emotional narration that reflects the story arc rather than just sentence-level emotion.

Future versions should explore multilingual TTS capability, particularly the addition of Tamil language support to broaden accessibility to Tamil-speaking Sri Lankan children. The Storypal application's use case strongly motivates multilingual deployment, as Sri Lanka's bilingual and trilingual context means that storybooks in Tamil are equally needed.

Finally, real-time streaming TTS — where audio playback begins before the full narration is synthesized — would significantly improve perceived responsiveness and user experience, particularly for longer storybooks. Integrating a streaming inference pipeline with Tacotron2 or a streaming-compatible alternative would be a valuable engineering milestone for future development.

3.4 Conclusion

The Storypal Text-to-Speech Synthesis module successfully demonstrates a functional and innovative approach to emotion-conditioned neural speech synthesis for inclusive storytelling. By integrating the Tacotron2 sequence-to-sequence model, WaveGlow flow-based vocoder, BERT emotion classifier, and a reference encoder for style conditioning, the system is capable of converting written storybook text into emotionally expressive audio narrations in real time.

Unlike traditional TTS systems that produce monotone, emotionally neutral output, this module enhances accessibility by preserving the emotional richness of story content in the synthesized narration. Visually impaired children interacting with the Storypal application receive an auditory storytelling experience that conveys not only the linguistic content of the story but also its emotional narrative arc — supporting comprehension, engagement, and emotional development.

The neutral speech synthesis component achieved a strong Mean Opinion Score (MOS) of 3.65, approaching human reference quality and confirming the suitability of the Tacotron2 + WaveGlow pipeline for high-quality neutral narration. The emotion-conditioned synthesis, while currently achieving a lower MOS of 1.65, establishes a clear proof-of-concept and identifies targeted improvement pathways for future development.

The BERT emotion classifier demonstrated robust performance with a macro-average F1-score of 0.84, successfully guiding the reference encoder's style conditioning for the majority of emotion categories. The system's usability evaluation yielded a SUS score of 81/100, confirming that the integrated audio narration experience is engaging and accessible for the target child user population.

The TTS module successfully fulfills its role within the broader Storypal system, providing the critical auditory bridge between text extracted from storybooks and the emotionally enriched listening experiences that make those stories come alive for children who cannot see the page. As a component of the larger collaborative research effort, it demonstrates how AI-driven speech synthesis can be purposefully applied to meaningful social challenges — advancing both the science of neural TTS and the practice of inclusive education technology.

Overall, the proposed TTS module represents a significant step toward inclusive and emotionally aware educational technology, contributing to better learning experiences for children with visual impairments and supporting the broader goal of accessible, emotionally resonant storytelling for all children.

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